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CMT Problem Set 4 Homework #4

Problem 4.1 (Structure factor for a tri-atomic basis). Consider a linear crystal with lattice constant a in which the basis consists of atom of species A at 0 and two atoms of species B at $1/3$ and $2/3$.

- Find the structure factor of such a crystal.
- X-rays of wavelength $\lambda (= a/4)$ irradiate this crystal at an oblique incidence (45°). Using the Ewald construction in the plane of incidence, show graphically, the diffraction directions and indicate the relative weights of the different diffracted intensities.
- State the physical significance of the vector $\mathbf{k}$ from which the extremity is on a plane passing through the origin of the reciprocal lattice.

Solution.

(a) Substitute $r_1 = 0, \frac{a}{3}, \frac{2a}{3}$ into the definition of the structure factor

$$S(h) = f_A e^{iG \cdot 0} + f_B e^{iG \cdot \frac{a}{3}} + f_B e^{iG \cdot \frac{2a}{3}} \quad \underline{G = 2\pi h/a} \quad f_A + f_B (e^{i\frac{2\pi h}{3}} + e^{i\frac{4\pi h}{3}})$$

Euler's formula $\begin{cases} f_A + 2f_B, & h=3n. \\ f_A - f_B, & \text{otherwise.} \end{cases}$

(b). The wave number

$$k = \frac{2\pi}{\lambda} = \frac{8\pi}{a} = 4b_0$$

where $b_0 = 2\pi/a$ is the reciprocal lattice vector. The incident wavevector is

$$\vec{k}_i = (k/\sqrt{2}, -k/\sqrt{2})$$

Then, the Ewald circle satisfies

$$(x + k/\sqrt{2})^2 + (y - k/\sqrt{2})^2 = k^2$$

and we can obtain two intersections on x-axis,

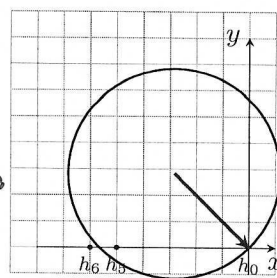
$$x_1 = 0, \quad x_2 = -\sqrt{2}k$$

obviously, $u_1 = 0, u_2 = -5.656$ ~~& Z~~

Since u_2 is not an integer, no other reciprocal ~~lattice~~ lattice points lies on the Ewald circle except the one on the original point. The intensity of each diffraction order is proportional to $|S(h)|^2$, i.e.,

$$I_h \propto |S(h)|^2 = \begin{cases} (f_A + 2f_B)^2, & h=3m \\ (f_A - f_B)^2, & \text{otherwise.} \end{cases}$$

(c) The tip of $\vec{k}$ lying on a plane XIII through the origin of the reciprocal lattice means the Bragg condition is satisfied for diffraction



Problem 4.2. The main experimental methods of X-ray diffraction are

- (a) the Laue method
- (b) the crystal rotation method
- (c) the powder method (or the Debye-Scherrer method)

Describe the corresponding experimental set-up and with the help of the Ewald construction in reciprocal space, deduce the observed diffraction patterns in each case.

Solution.

(a) Laue method

Setup. Stationary single crystal. polychromatic X-rays

Ewald & Pattern. A continuous range of Ewald sphere radii, which is due to varying λ , means many reciprocal lattice points lie on some sphere. This produces a pattern of sharp spots, each from a different (hkl) plane and wavelength. The spot arrangement directly reveals the crystal's symmetry.

(b) Crystal rotation method.

Setup. Rotating single crystal, monochromatic X-rays.

Ewald & Pattern. The fixed Ewald sphere is fixed and intersected by rotating reciprocal lattice points. This generates discrete spots or arcs on the detector. Measuring the rotation angles and intensities of these spots allows for the determination of the unit cell and atomic sphere.

(c) Powder method.

Setup. Powdered sample with random crystallites, monochromatic X-rays

Ewald & Pattern. Each set of (hkl) planes forms a spherical shell in reciprocal space. The Ewald sphere's intersection with these shells creates cones of diffracted beams, i.e., Debye cones, which appear as characteristic rings on the detector. The ring positions, i.e., 2θ angles, provide d -spacings for phase identification and lattice parameter calculation.